

MD SHAKHRUL IMAN SIAM

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RESEARCH INTEREST

Multi-modal Large language Models, Audio and Speech signal processing, Egocentric Video processing, Generative AI for Healthcare.

WORK EXPERIENCE

- Microsoft**
 - Research Intern**Research Focus: ASR Privacy, Multimodal LLM, Neural Audio Codec**
 - **DECAF: Disentangling Encoder for Compressed Audio Features**
 - Designed a Neural Audio Codec that disentangles speaker identity from voice data and transmits with a very low bitrate.
 - Achieved 13.5% improvement of Equal Error Rate compared to state-of-the-art Voice conversion system with only 6.6% degradation of relative WER compared to original audio.
 - Fine-tune an LLM-based Automatic Speech Recognition System that uses the transmitted codes directly as input without the need to reconstruct to the audio waveform.

USA
June 2025 - August 2025
- The Ohio State University**
 - Graduate Research Associate, OSU AIoT and Machine Learning Systems Lab 🌐**Advisor: Prof. Mi Zhang**
 - **Multimodal Audio-language Foundation Model for Respiratory Health Diagnosis**
ACL 2025 Main Conference
 - Developed a Multimodal Foundation model that will capture sounds from a smartphone microphone and perform several downstream tasks like disease (Covid-19, TB, COPD) detection, generating health reports, and Question-answering.
 - Utilize Large Language Models with PEFT and LoRA techniques to fine-tune multimodal models from cough sounds and self-reported patient text metadata.
 - **A large-scale multi-modal reading dataset of 100 hours of egocentric videos**
NeurIPS 2025 Poster
 - A first-of-its-kind dataset that enables egocentric and spatial AI on the next-generation Meta AI glasses.
 - Developed a transformer-based model that detects whether and when a user is reading using different input modalities (e.g., egocentric video, eye gaze, IMU)
 - Developed a benchmark using both gaze information and RGB image data to classify reading and non-reading activities.
 - **Gaze-guided Keyframe Selection for Egocentric Long Video Understanding**
 - Developed a training-free keyframe selection framework for long egocentric video understanding, achieving up to 15.7% accuracy improvement over SOTA sampling methods across various Vision Language Models(VLMs).
 - Curated a novel benchmark of 3,174 multiple-choice VQA samples with gaze annotations.
 - Designed a cross-modal query-frame retrieval pipeline and cost-aware Maximal Marginal Relevance (MMR) selection for keyframe selection.
 - **Visually-Grounded Entity Graph for Egocentric Long Video Question Answering**
 - Designed VEG-QA, a training-free multi-agent system for long-form egocentric video QA that unifies visual embeddings and structured graph representations.
 - Constructed a visually-grounded entity graph over week-long egocentric streams with typed nodes (episodes, persons, objects, locations, activities) storing CLIP embeddings alongside textual attributes.
 - Outperformed SOTA agentic baselines on EgoLife QA by replacing disconnected visual stores with unified graph-native visual grounding.

Columbus, Ohio, USA
August 2023 - Present
- ACI Limited**
 - Machine Learning Engineer**Domain Experience: Computer Vision, Speech Recognition, Predictive analysis, NLP**
 - **Person re-identification and tracking using multiple CCTV camera feeds.** 🌐
 - Developed a system to re-identify, track, and monitor customers' movement in Retail outlets from CCTV camera.
 - Person Detection using YOLOv5 model trained on a custom dataset, and Re-identification using embedding generated by a Resnet-50 model trained on triplet loss.
 - Developed a 2D projection algorithm to map each persons' position from camera image to floorplan.
 - **Facial Recognition based automatic attendance system.** 🌐
 - Implemented face detection using Retinaface and face recognition using Arcface.
 - Converted the model to Tflite to improve inference speed by 4×.

Dhaka, Bangladesh
October 2021 - July 2023

- Designed the database (MySQL) for daily attendance entry and built a dashboard with Django.
- **Medicine name recognition system from handwritten prescription.** [↗](#)
 - Developed a ROI Extraction model by using LayoutLM.
 - Developed OCR models using EasyOCR, PyTesseract, and PaddleOCR to recognize medicine names.
- **Voice Controlled Conversational Chatbot.** [↗](#)
 - Implemented a streaming ASR model and a Text-to-speech model for automated speech recognition of users.
 - Designed a sliding window technique to provide the ASR model with real-time streaming capabilities.

EDUCATION

- **The Ohio State University** Columbus, Ohio, USA
Ph.D. in Computer Science and Engineering (CSE) 08/2023 - 12/2027 (Expected)
Selected Courses: Artificial Intelligence, Speech and Language Processing, Neural Networks
- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
BSc. in Electrical and Electronic Engineering (EEE) 02/2016 - 02/2021
Selected Courses: Digital Signal processing, Microcontroller, Wireless communication

PUBLICATIONS

Published

- **Shakhrul Iman Siam**, Tiantian Feng, Jiankun Zhang, Shrikanth Narayanan and Mi Zhang. “RespiraMFM: A Multimodal Foundation Model with Contrastive Audio-Language Alignment for Respiratory Disease Identification.” (ACL 2026)
- Samiul Alam, **Shakhrul Iman Siam**, Mi Zhang “MemoLens: Empowering Augmented Reality Glasses with Super Memory” (Mobisys 2026).
- Charig Yang, Samiul Alam, **Shakhrul Iman Siam**, Michael J. Proulx, Lambert Mathias, Kiran Somasundaram, James Fort, Omkar Parkhi, Yuheng Ren, Mi Zhang, Yuning Chai, Richard Newcombe, Hyo Jin Kim. “Reading Recognition in the Wild” (NeurIPS 2025). [\[Project Link\]](#)
- **Shakhrul Iman Siam**, Hyunho Ahn, Li Liu, Samiul Alam, Hui Shen, Zhichao Cao, Ness Shroff, Bhaskar Krishnamachari, Mani Srivastava, and Mi Zhang. “Artificial Intelligence of Things: A Survey” ACM Transactions on Sensor Networks (TOSN 2024). [Paper Link](#)
- Chenning Li, Yidong Ren, Shuai Tong, **Shakhrul Iman Siam**, Mi Zhang, Jiliang Wang, Yunhao Liu, and Zhichao Cao. “ChirpTransformer: Versatile LoRa Encoding for Low-power Wide-area IoT.” In Proceedings of the 22nd Annual International Conference on Mobile Systems, Applications and Services (Mobisys 2024). [\[Paper Link\]](#)
- **Md Shakhrul Iman Siam**, and Subrata Biswas. “A Deep Learning Based Person Detection and Heatmap Generation Technique with a Multi-Camera System.” In 2022 12th International Conference on Electrical and Computer Engineering (ICECE 2022).
- **Md Shakhrul Iman Siam**, Md Saiful Bari Siddiqui, Mushfiqul Abedin, and Mohammed Imamul Hassan Bhuiyan. “Bioradiolocation-Based Multi-Class Sleep Stage Classification Using Time and Frequency Features with Random Forest Classifier.” In 2022 12th International Conference on Electrical and Computer Engineering (ICECE 2022).
- KM Naimul Hassan, Subrata Kumar Biswas, Md Shakil Anwar, **Md Shakhrul Iman Siam**, and Celia Shahnaz. “A Dual-Purpose Refreshable Braille Display Based on Real Time Object Detection and Optical Character Recognition.” In 2019 IEEE International Conference on Signal Processing, Information, Communication & Systems (SPICSCON 2019).

Under Review

- **Shakhrul Iman Siam**, Dushyant Sharma, Stanislav Yu. Kruchinin, and Peter Skala. “DECAF: Disentangling Encoder for Compressed Audio Features”.
- **Shakhrul Iman Siam**, Samiul Alam, Mi Zhang “Gaze-guided Key-frame selection in Egocentric Long Video Understanding task”.

SKILLS SUMMARY

Programming Languages	Python, C, C++, JavaScript, MATLAB, Assembly(ARM), Unix, Shell Scripts
Machine Learning Domain	Computer Vision, Generative AI, Foundation Model, Face-Recognition, Predictive Analysis, Recommendation System, Chatbot, Reinforcement Learning
ML-DL Frameworks	Pytorch, TensorFlow, Keras, Scikit-learn, NLTK, Pandas, OpenCV, Numpy, Unity
LLM Frameworks	MLC-LLM, DeepSpeed
Backend Frameworks	Django, Django Rest Framework
Databases	MySQL, Oracle
Web Development	HTML, CSS, BootStrap
Cloud & Containerization	AWS, Docker, Kubernetes
Hardware & IoT	Arduino, Raspberry pi, Microcontroller, Meta Aria Glasses

SELECTED PROJECTS

- **Generating Customer Heatmap of a retail outlet from CCTV camera :** Using Yolov5 object detection model, Homography transformation and Kernel density estimator to generate customer density map on a floorplan of a retail outlet using video taken from CCTV camera. [↗](#)
- **Blind vision Assistance :** Prototype of a low-cost mobile device, that can assist a visually disabled person to read, recognize person, and detect objects using voice command. There are three main features of this system- Object detection, Face Recognition, and optical character recognition. [↗](#)
- **Real-time vehicle detection and tracking at junction using a fisheye camera :** The Video and Image Processing cup 2020 challenge focuses on fisheye cameras mounted into street lamps at junctions and vehicle detection and tracking to be used for a junction management system to optimize the flow of traffic and synchronize with other junctions to obtain bottleneck performances throughout the city. [↗](#)
- **Open Source Contribution in developing a pypi package:** A Python library for offline reverse geocoding. Reverse Geocode BD takes a (latitude , longitude) coordinate and returns the Division, District, Upazila and Thana of any location in Bangladesh. [↗](#)
- **Unsupervised Synthetic/Fake Speech Detection.** Our research introduces a novel unsupervised model, which we have termed "Anocoder". This model employs a one-dimensional convolutional autoencoder and is designed to differentiate between authentic and synthetic or fabricated speech.
- **Web Scrapping and Sentiment Analysis from Social Media post:** Using Selenium for web scrapping and analysis of social media and online newspaper posts for sentiment analysis. Tech: Selenium, BeautifulSoup

HONORS AND AWARDS

- First Runner up at Robi **Datathon 2.0** (2022)
- 5th at **IEEE Video and Image Processing Cup** (2020)
- Champion of Bangladesh Section and World Finalist, **IEEE YESIST12 Innovation Challenge** (2019)
- 1st Runner up, BUET CSE Day **Math Olympiad** (2017)
- 1st Runner up, BAS Divisional **Science Olympiad** (2015)
- Champion , Dutch Bangla Bank Prothom Alo Regional **Math Olympiad** (2015)
- 2nd Runner Up , Dutch Bangla Bank Prothom Alo Regional **Math Olympiad** (2014)
- Champion , Dutch Bangla Bank Prothom Alo Regional **Math Olympiad** (2011)

REFERENCE

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